Chapter 3 RATIO AND PROPORTION

Learning Objectives

- To understand the concept of ratio.
- To use ratio notation and simplify ratios.
- To divide a quantity into two parts for a given ratio.
- To recognise the relationship between ratio and proportion.
- To use the unitary method and solve simple ratio problems.

Recap

1. Which of the following fractions is not a proper fraction?

   (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{5}{10}$ (d) $\frac{10}{5}$

2. The equivalent fraction of $\frac{1}{7}$ is _____.

   (a) $\frac{2}{15}$ (b) $\frac{1}{49}$ (c) $\frac{7}{49}$ (d) $\frac{100}{7}$

3. Write > ,< or = in the box.

   (i) $\frac{5}{8}$ ☐ $\frac{1}{10}$ (ii) $\frac{9}{12}$ ☐ $\frac{3}{4}$

4. Anban says that $\frac{2}{6}$th of the group of triangles given below are blue. Is he correct?

5. Joseph has a flower garden. Draw a picture which shows that $\frac{2}{10}$ th of the flowers are red and the rest of them are yellow.

6. Malarkodi has 10 oranges. If she ate 4 oranges, what fraction of oranges was not eaten by her?

7. After sowing seeds on day one, Muthu observes the growth of two plants and records it. In 10 days, if the first plant grew $\frac{1}{4}$ th of an inch and the second plant grew $\frac{3}{8}$ th of an inch, then which plant grew more?

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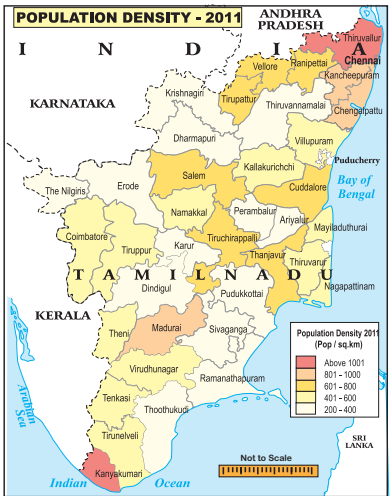
3.1 Introduction

In our daily life, we handle lots of situations where we compare quantities. Comparison of our heights, weights, marks secured in examinations, speeds of vehicles, distances travelled, auto fare to taxi fare, bank balances at different periods of time and many more things are done. Comparison is usually between quantities of the same kind and not of different kind. It will not be meaningful to compare the height of a person with the age of another person. Also, we need a standard measure for comparison.

This sort of comparison by expressing one quantity as the number of times the other is called a '*Ratio*'.

MATHEMATICS ALIVE – RATIO IN REAL LIFE



555 persons to one square k.m.

Male		Female	
Height	Ideal Weight	Height	Ideal Weight
4' 6"	28 - 35 Kg.	4' 6"	28 - 35 Kg.
4' 7"	30 - 39 Kg.	4' 7"	30 - 37 Kg.
4' 8"	33 - 40 Kg.	4' 8"	32 - 40 Kg.
4' 9"	35 - 44 Kg.	4' 9"	35 - 42 Kg.
4' 10"	38 - 46 Kg.	4' 10"	36 - 45 Kg.
4' 11"	40 - 50 Kg.	4' 11"	39 - 47 Kg.
5' 0"	43 - 53 Kg.	5' 0"	40 - 50 Kg.
5' 1"	45 - 55 Kg.	5' 1"	43 - 52 Kg.
5' 2"	48 - 59 Kg.	5' 2"	45 - 55 Kg.
5' 3"	50 - 61 Kg.	5' 3"	47 - 57 Kg.
5' 4"	53 - 65 Kg.	5' 4"	49 - 60 Kg.
5' 5"	55 - 68 Kg.	5' 5"	51 - 62 Kg.
5' 6"	58 - 70 Kg.	5' 6"	53 - 65 Kg.
5' 7"	60 - 74 Kg.	5' 7"	55 - 67 Kg.
5' 8"	63 - 76 Kg.	5' 8"	57 - 70 Kg.
5' 9"	65 - 80 Kg.	5' 9"	59 - 72 Kg.
5' 10"	67 - 83 Kg.	5' 10"	61 - 75 Kg.
5' 11"	70 - 85 Kg.	5' 11"	63 - 77 Kg.
6' 0"	72 - 89 Kg.	6' 0"	65 - 80 Kg.

Comparison of Height and Weight by using ratio

3.2 Ratio

Think about this Situation

Let us consider a situation of cooking rice for two persons. The quantity of rice required for two persons is one cup. To cook every one cup of rice, we need to add two cups of water. Assuming that 8 more guests join for lunch, will the use of ratio help us in handing this situation?

The number of cups of rice and water required are given below.

DO YOU KNOW?

It is possible to trace the origin of the word "ratio" to the Ancient Greek Medieval. Writers used the word proprotio ("proportion") to indicate ratio and proportionalities ("proprotionality") for the equality of ratios. Early translators rendered this into Latin as ratio ("reason"; as in the word "rational")

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Number of cups of rice	1	2	3	4	5
Number of cups of water (or) No. of persons	2	4	6	8	10

In all the cases, the number of cups of water (or) the number of persons is 2 times the number of cups of rice. So, we write

Number of cups of rice : Number of cups of water (or) the number of persons = 1 : 2

Such comparison is called as a Ratio.

NOTE

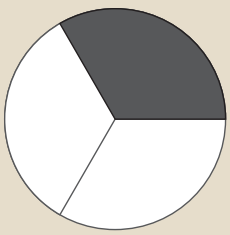
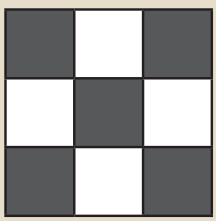
- A ratio is a comparison of two quantities with same units.
- If 'a' and 'b' are two different quantities with same unit then, we can write the ratio as a:b (read as a is to b).
- A ratio can be written as a fraction; ratios are mostly written in the simplest form.
- In the above example, the ratio of rice to water in terms of the number of cups can be written in three different ways as 1 : 2 or $\frac{1}{2}$ or 1 to 2 .

TRY THESE

1. Write the ratio of red tiles to blue tiles and yellow tiles to red tiles.

2. Write the ratio of blue tiles to that of red tiles and red tiles to that of total tiles.

3. Write the ratio of shaded portion to the unshaded portions in the following shapes.

	
Ratio:_____	Ratio:_____

3.2.1 Properties of Ratio

- A ratio has no unit. It is a number. For example, the ratio of 8 km to 4 km is written as 8 : 4 = 2 : 1 and not 2 km : 1 km
- The two quantities in a ratio should be of the *same* unit. The ratio of 4 km to 400 m is expressed as (4 × 1000) : 400 = 4000 : 400 = 10 : 1

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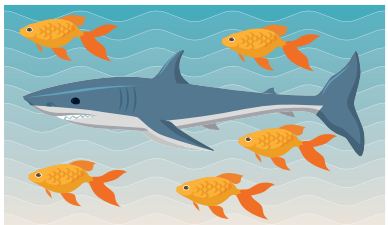
- Each number of the ratio is called *a term*.
- Order of the terms in a ratio cannot be reversed.

A few examples are given below.

 (a)	(b)
Ratio of the number of small fish to the number of big fish is 5 : 1	Ratio of the number of boys to girls in above figure is 5 : 4

In the above example, the ratio of the number of small fish to the number of big fish is 5 : 1. The same information cannot be written as 1 : 5 and so, 5 : 1 and 1 : 5 are not the same.

If in a class, there are 12 boys and 12 girls, then the ratio of number of boys to the number of girls is expressed as 12 : 12 which is the same as 1 : 1.

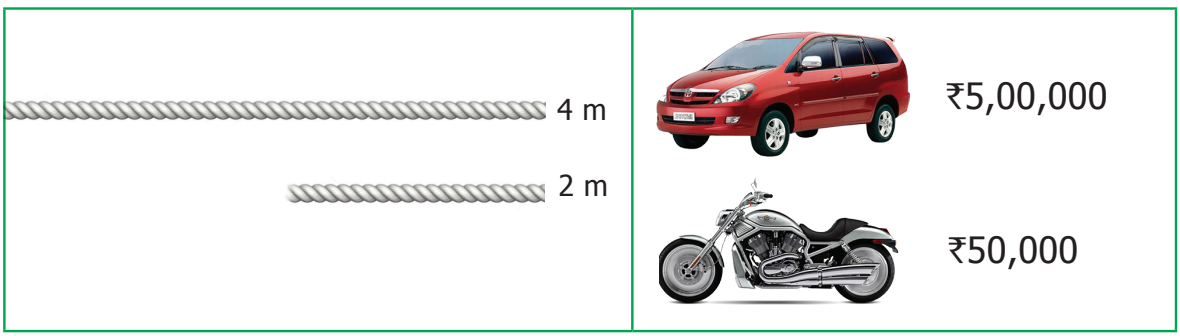
TRY THESE

If the given quantity is in the same unit, put ✓ otherwise put X in the table below.

Sl.No	Quantity	Put ✓ or X
1	5 cm and 100 cm	
2	₹ 5 and 50 oranges	
3	2 m and 75 ml	
4	7 km and 700 m	
5	3 kg of potatoes and 2 kg onions	
6	10 cm and 32 pencils	

3.2.2 Ratios in simplest form

Think about these situations



(a) Fig. 3.1 (b)

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1. The larger rope is 4 m long and the smaller rope is 2 m long. This is expressed in the form of ratio as 4 : 2 and the simplest form of ratio of the larger rope to the smaller rope is 2 : 1 (See Fig. 3.1 (a))
2. The cost of a car is ₹ 5,00,000 and the cost of a motorbike is ₹ 50,000. This is expressed as 500000 : 50000 = 50 : 5 and the simplest form of ratio of the car to the motorbike is 10 : 1 (See Fig. 3.1 (b))

3.2.3 Simplifying ratios of same unit

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

Example 3.1

Simplify the ratio 20 : 5.

Solution

Step 1 : Write the ratio in fraction form as $\frac{20}{5}$.

Step 2 : Divide numerator and denominator by 5. That is, $\frac{20 \div 5}{5 \div 5} = \frac{4}{1} = 4 : 1$

This is the ratio in the simplest form.

Example 3.2

Find the ratio of 500 g to 250 g.

Solution

500 g to 250 g = 500 : 250 $\Rightarrow \frac{500}{250} = \frac{500 \div 250}{250 \div 250} = \frac{2}{1} = 2 : 1$

This is the ratio in the simplest form.

Example 3.3

Madhavi and Anbu bought two tables for ₹ 750 and ₹ 900 respectively. What is the ratio of the prices of tables bought by Anbu and Madhavi?

Solution

The ratio of the price of tables bought by Anbu and Madhavi

$= 900:750 = \frac{900}{750} \Rightarrow \frac{900 \div 150}{750 \div 150} = \frac{6}{5} = 6:5$. This is the ratio in the simplest form.

3.2.4 Simplifying ratios of different units

1 hour = 60 minutes

20 x 1 = 20

20 x 2 = 40

20 x 3 = 60

Example 3.4

What is the ratio of 40 minutes to 1 hour?

Solution

Step 1 : Express the quantity in the same unit. (Hint : 1 Hour = 60 minutes)

Step 2 : Now, the ratio of 40 minutes to 60 minutes is $40:60 \Rightarrow \frac{40}{60} = \frac{40 \div 20}{60 \div 20} = \frac{2}{3} = 2:3$

This is the ratio in the simplest form.

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TRY THESE

Write the ratios in the simplest form and fill in the table.

Sl. No.	Quantity	Ratio Form	Fraction Form	Dividing by a common number	Simplest form of Ratio
1	Ratio of 15 girls to 10 boys	15 : 10	$\frac{15}{10}$	$\frac{15 \div 5}{10 \div 5} = \frac{3}{2}$	3 : 2
2	Ratio of 1m 25 cm to 2 m	125 : 200 (1m = 100cm)	$\frac{125}{200}$		
3	Ratio of 3 Kg to 750 g	3000 : 750 (1Kg = 1000g)			
4	Ratio of 70 minutes to 30 minutes				

Exercise 3.1

1. Fill in the blanks.
   (i) Ratio of ₹3 to ₹5 = _____.
   (ii) Ratio of 3 m to 200 c.m = _____.
   (iii) Ratio of 5 k.m 400 m to 6 k.m = _____.
   (iv) Ratio of 75 paise to ₹2 = _____.
2. Say whether the following statements are True or False.
   (i) The ratio of 130 c.m to 1 m is 13 : 10.
   (ii) One of the terms in a ratio cannot be 1.
3. Find the simplified form of the following ratios.
   (i) 15 : 20 (ii) 32 : 24 (iii) 7 : 15 (iv) 12 : 27 (v) 75 : 100
4. Akilan walks 10 k.m in an hour while Selvi walks 6 km in an hour. Find the simplest ratio of the distance covered by Akilan to that of Selvi.
5. The cost of parking a bicycle is ₹5 and the cost of parking a scooter is ₹15. Find the simplest ratio of the parking cost of a bicycle to that of a scooter.
6. Out of 50 students in a class, 30 are boys. Find the ratio of
   (i) number of boys to the number of girls.
   (ii) number of girls to the total number of students.
   (iii) number of boys to the total number of students.

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Objective Type Questions

7. The ratio of ₹1 to 20 paise is _____.

   (a) 1 : 5 (b) 1 : 2 (c) 2 : 1 (d) 5 : 1

8. The ratio of 1 *l* to 50 *ml* is _____.

   (a) 1 : 5 (b) 1 : 20 (c) 20 : 1 (d) 5 : 1

9. The length and breadth of a window are in 1m and 70 cm respectively. The ratio of the length to the breadth is _____.

   (a) 1 : 7 (b) 7 : 1 (c) 7 : 10 (d) 10 : 7

10. The ratio of the number of sides of a triangle to the number of sides of a rectangle is

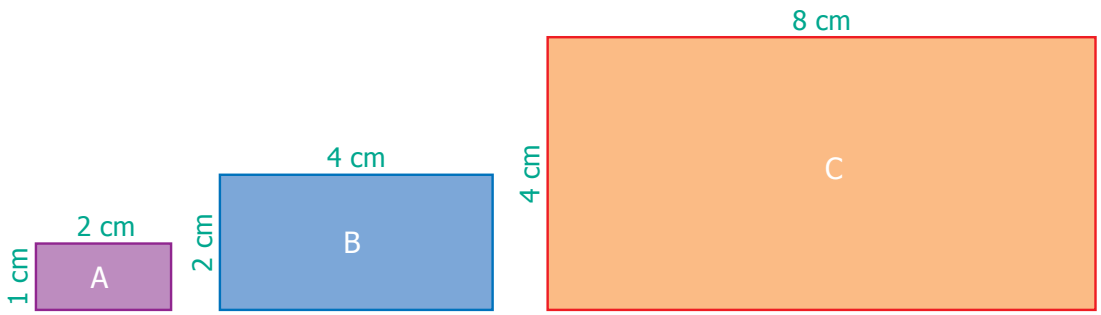
    (a) 4 : 3 (b) 3 : 4 (c) 3 : 5 (d) 3 : 2

11. If Azhagan is 50 years old and his son is 10 years old then the simplest ratio between the age of Azhagan to his son is

    (a) 10 : 50 (b) 50 : 10 (c) 5 : 1 (d) 1 : 5

3.2.5 Equivalent Ratios

We can get equivalent ratios by multiplying or dividing the numerator and denominator by a common number. This is clear from the following example. Let us find the ratio between breadth and length of the following rectangles given below.



- Ratio of breadth to length of **rectangle A** is **1 : 2** (already in simplest form)
- Ratio of breadth to length of **rectangle B** is **2 : 4** (simplest form is 1 : 2)
- Ratio of breadth to length of **rectangle C** is **4 : 8** (simplest form is 1 : 2)
- Thus, the ratios of breadth and length of rectangles **A**, **B** and **C** are said to be *equivalent ratios*.
- That is, the ratios **1 : 2** = **2 : 4** = **4 : 8** are *equivalent*.

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TRY THESE

1. For the given ratios, find two equivalent ratios and complete the table.

	Ratio	Fraction Form	Equivalent ratio
(i)	1 : 3	$\frac{1}{3}$	$\frac{1}{3} \times \frac{2}{2} = \frac{2}{6} = 2:6$ and $\frac{1}{3} \times \frac{3}{3} = \frac{3}{9} = 3:9$
(ii)	3 : 7	$\frac{3}{7}$	
(iii)	5 : 8	$\frac{5}{8}$	

2. Write three equivalent ratios and fill in the boxes.

	Ratio	Equivalent Ratios		
(i)	4 : 5	8 : ☐	☐ : 50	12 : ☐
(ii)	7 : 2	☐ : 10	14 : ☐	49 : ☐
(iii)	8 : 5	32 : ☐	☐ : 50	16 : ☐

3. For the given ratios, find their simplest form and complete the table.

	Ratio	Fraction Form	Simplest form
(i)	5 : 60	$\frac{5}{60}$	$\frac{5 \div 5}{60 \div 5} = \frac{1}{12} = 1:12$
(ii)	4000 : 6000	$\frac{4000}{6000}$	
(iii)	1100 : 5500		

3.2.6 Comparison of Ratios

Consider the following situations.

Situation 1

In the Idly Batter, the ratio of black gram to idly rice is 1 : 4

In the Idly Batter, the ratio of black gram to idly rice is 1 : 3

(a) Fig. 3.2 (b)

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Can you find which ratio is greater in Fig. 3.2?

Express ratios as a fraction and then find the equivalent fractions, until the denominators are the same, and compare the fractions with common denominators. This is done as follows :

Idly Batter Ratio - (a)	Idly Batter Ratio - (b)
$\frac{1}{4} \times \frac{2}{2} = \frac{2}{8}$	$\frac{1}{3} \times \frac{2}{2} = \frac{2}{6}$
$\frac{1}{4} \times \frac{3}{3} = \frac{3}{12}$	$\frac{1}{3} \times \frac{3}{3} = \frac{3}{9}$
$\frac{1}{4} \times \frac{4}{4} = \frac{4}{16}$	$\frac{1}{3} \times \frac{4}{4} = \frac{4}{12}$

Comparing the equivalent ratios, $\frac{4}{12}$ & $\frac{3}{12}$, we can conclude that 1 : 3 is greater than 1 : 4

Situation 2

Let us consider another situation. For example, if a thread of 5 m is cut at 3 m, then the length of two pieces are 3 m and 2 m and the ratio of the two pieces is 3 : 2. From this we say that, a ratio '$a : b$' is said to have a total of '$a+b$' parts in it.

Example 3.5

Kumaran has ₹600 and wants to divide it between Vimala and Yazhini in the ratio 2 : 3. Who will get more and how much?

Solution

Divide the whole money into 2 + 3 = 5 equal parts then, Vimala gets 2 parts out of 5 parts and Yazhini gets 3 parts out of 5 parts.

Amount Vimala gets = ₹ 600 × $\frac{2}{5}$ = ₹ 240

Amount Yazhini gets = ₹ 600 × $\frac{3}{5}$ = ₹ 360

Vimala received ₹ 240 and Yazhini gets ₹ 360, which is ₹ 120 more than that of Vimala.

Exercise 3.2

W39CV

1. Fill in the blanks for the given equivalent ratios.

   (i) 3 : 5 = 9 : ___ (ii) 4 : 5 = ___ : 10 (iii) 6 : ___ = 1 : 2

2. Complete the table.

(i)

Feet	1	2	3	?
Inch	12	24	?	72

(ii)

Days	28	21	?	63
Weeks	4	3	2	?

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3. Say True or False.

(i) 5 : 7 is equivalent to 21 : 15

(ii) If 40 is divided in the ratio 3 : 2, then the larger part is 24.

4. Give two equivalent ratios for each of the following.

(i) 3 : 2 (ii) 1 : 6 (iii) 5 : 4

5. Which of the two ratios is larger?

(i) 4 : 5 or 8 : 15 (ii) 3 : 4 or 7 : 8 (iii) 1 : 2 or 2 : 1

6. Divide the numbers given below in the required ratio.

(i) 20 in the ratio 3 : 2 (ii) 27 in the ratio 4 : 5 (iii) 40 in the ratio 6 : 14

7. In a family, the amount spent in a month for buying Provisions and Vegetables are in the ratio 3 : 2. If the allotted amount is ₹4000, then what will be the amount spent for (i) Provisions and (ii) Vegetables?

8. A line segment 63 cm long is to be divided into two parts in the ratio 3 : 4. Find the length of each part.

Objective Type Questions

9. If 2 : 3 and 4 : ___ are equivalent ratios, then the missing term is

(a) 6 (b) 2 (c) 4 (d) 3

10. An equivalent ratio of 4 : 7 is

(a) 1 : 3 (b) 8 : 15 (c) 14 : 8 (d) 12 : 21

11. Which is not an equivalent ratio of $\frac{16}{24}$?

(a) $\frac{6}{9}$ (b) $\frac{12}{18}$ (c) $\frac{10}{15}$ (d) $\frac{20}{28}$

12. If ₹1600 is divided among A and B in the ratio 3 : 5 then, B's share is

(a) ₹ 480 (b) ₹ 800 (c) ₹ 1000 (d) ₹ 200

3.3 Proportion

When two ratios are equal $\left(\frac{a}{b} = \frac{c}{d}\right)$, we say that the ratios are in Proportion.This is denoted as $a : b :: c : d$ and it is read as 'a is to b as c is to d'. The following situations explain about proportion.

Situation 1

The Teacher said to the students, "You can do a maximum of 4 projects in Mathematics. You will get 5 as internal marks for each project that you do". Kamala asked, "Teacher, What if I do 2 or 3 or 4 projects?" The teacher replied, "For 2 projects you will get 10 marks, for 3 projects you will get 15 marks and for 4 projects you will get 20 marks".

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